Chapter 5

Square Root and Cube Root

5.1 Square

If a number is multiplied with itself, then the result of this multiplication is called the square of that number. *For example:*

- Square of $6 = 6 \times 6 = 36$
- Square of $12 = 12 \times 12 = 144$
- Square of $100 = 100 \times 100 = 10000$

Methods to Find Square

Different methods to calculate the square of a number are as follows

Multiplication Method

In this method, the square of any 2-digit number can be calculated by the following given steps.

Step I: Square the unit's digit

Step II: 2× Ten's digit × Unit's digit + Carry

Step III: 2(Ten's digit) + Carry from step II

Step IV: Now arrange the numbers first write step III number, then step II and at unit's place step I.

For example:

Find the square of 74

Step I	$4^2 = 16$	{Carry = 1}
Step II	$(2 \times 7 \times 4) + 1 = 57$	{Carry = 5}
Step III	$7^2 + 5 = 49 + 5 = 54$	
Step IV	$74^2 = 5476$	

Algebraic Method

To calculate square by this method, two formulae are used.

★ $(a+b)^2 = a^2 + b^2 + 2ab$

★ $(a-b)^2 = a^2 + b^2 - 2ab$

For example: The square of 34 is

$$(34)^2 = (30+4)^2 = (30)^2 + (4)^2 + 2 \times 30 \times 4$$
$$= 900 + 16 + 240 = 1156$$

5.2 Square of Decimal Numbers

To find the square of any decimal number, write the square of the number ignoring the decimal and then place the decimal twice the place of the original number starting from unit's place.
For example: The square of 3.5 is as follows

$$35^2 = 1225$$

Here, the decimal is after one-digit in 3.5. Hence, the decimal will be placed twice the place of original number in the result.

$$(3.5)^2 = 12.25$$

5.3 Square Root

The square root of a number is that number, the square of which is equal to the given number. There are two types of square roots of a number, positive and negative. It is denoted by the sign $\sqrt{}$.
For example: 49 has two square roots 7 and - 7, because $(7)^2 = 49$ and $(-7)^2 = 49$. Hence, we can write $\sqrt{49} = \pm 7$.

Methods to Find Square Root

Different methods to calculate the square root of a number are as follows:

Prime Factorisation Method

This method has the following steps:

Step I Express the given number as the product of prime factors.

Step II Arrange the factors in pairs of same prime numbers.

Step III Take the product of these prime factors taking one out of every pair of the same primes. This product gives us the square root of the given number

Ex.1 Find the square root of 1089.

Sol. Prime Factors of

$$1089 = 11 \times 11 \times 3 \times 3$$

$$\Longrightarrow \quad \sqrt{1089} = \sqrt{11 \times 11 \times 3 \times 3}$$

Now, taking one number from each pair and multiplying them, we get

$$\sqrt{1089} = 11 \times 3 = 33$$

11	1089
11	99
3	9
3	3
	1

2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

Ex. 2 Find the square root of 1024.

Sol. Prime factors of

$$1024 = 2\times2\times2\times2\times2\times2\times2\times2\times2\times2$$

$$\Longrightarrow \quad \sqrt{1024} = \sqrt{2\times2\times2\times2\times2\times2\times2\times2\times2\times2}$$

Now, taking one number from each pair and multiplying them, we get

$$\sqrt{1024} = 2\times2\times2\times2\times2 = 32$$

Division Method

If it is not easy to evaluate square root using prime factorisation method, then we use division method.
The steps of this method can be easily understood with the help of following examples.
Ex. 3 Find the square root of 18769.
Sol.

Step I In the given number, mark off the digits in pairs starting from the unit digit. Each pair and the remaining one-digit (if any) is called a period.

Step II Now, 1=1; On subtracting, we get 0 (zero) as remainder.

Step III Bring down the next period, i.e. 87. Now, the trial divisor is $1\times2 = 2$ and trial dividend is 87. So, we take 23 as divisor and put 3 as quotient. The remainder is 18 now.

Step IV Bring down the next period, which is 69. Now, trial divisor is $13\times2 = 26$ and trial dividend is 1869. So, we take 267 as dividend and 7 as quotient. The remainder is 0.

Step V The process (processes like III and IV) goes on till all the periods (pairs) come to an end and we get remainder as 0 (zero) now. Hence, the required square root= 137

	137
1	1 87 69
	1 ↓
23	87
	69 ↓
267	1869
	1869
	0

	389
3	15 13 21
	9 ↓
68	613
	544 ↓
769	6921
	6921
	0

Ex. 4 What is the square root of 151321?
Sol. From above figure, Required square root = 389

5.4 Properties of Squares and Square Roots

- The difference of squares of two consecutive numbers will always be equal to the sum of the number i.e., $(a^2 - b^2) = (a + b)(a - b)$. Here, $a > b$ and (a, b) being consecutive $(a—b) = 1$.

  For example: If $a = 12$ and $b = 11$,
  then $(12^2 - 11^2) = (12 + 11)(12 - 11) = 23$

- If the square of any number ends with 1, then its square root will end with 1 or 9.
- If the square of any number ends with 4, then its square root will end with 2 or 8.
- If the square of any number ends with 5, then its square root will end with 5.
- If the square of any number ends with 6, then its square root will end with 7 or 6.
- If the square of any number ends with 9, then its square root will end with 3 or 7.
- The square of any number always ends with 0,1,4, 5,6 or 9 but will never end with 2,3, 7 or 8,
- Square root of negative number is imaginary.

Important Relations

1. $\sqrt{?} = y$. The required number $= y^2$
2. $\sqrt{a^2 \times b^2} = a \times b$
3. $\sqrt{a^3 \times b^3} = a \times b\sqrt{a \times b}$
4. $\sqrt{a^4 \times b^4 \times c^4} = a^2 \times b^2 \times c^2$
5. $\sqrt{a^n \times b^m} = a^{n/2} \times b^{m/2}$
6. $\sqrt{x} \times \sqrt{y} = \sqrt{xy}$

7. $\frac{\sqrt{x}}{\sqrt{y}} = \sqrt{\frac{x}{y}}$

Square Root of Decimal Numbers

If in a given decimal number, the number of digits after decimal are not even, then we put a 0 (zero) at the extreme right, So that these are even number of digits after the decimal point. Now, periods are marked as marked in previous explanation starting from right hand side before the decimal point and from the left hand after the decimal digit.
For example: 156.694
There are odd number of digits after decimal. So, we put a zero after the digit, so that there are even digits after the decimal 156.6940
Now, periods are marked as

Direction of marking
$\longleftarrow$ $\longrightarrow$
$156 \cdot 6940$

After the periods are marked, then previous method is used to find the square root.
Ex.5 Find the square root of 147.1369.
Sol. Here, 147.1369 contains even digits after decimal, so there is no need to add zero after the the last digit, now period are marked as 147.1369

$\therefore$ Required Number is 12.13

	12.13
1	1 47 . 13 69
	1 ↓
22	47
	44 ↓
241	313
	241 ↓
2423	7269
	7269
	0

	12.231
1	1 49 . 59 73 61
	1 ↓
22	49
	44 ↓
242	559
	484 ↓
2443	7573
	7329 ↓
24461	24461
	24461
	0

Ex.6 Find the square root of 149.597361
Sol.

$\therefore$ Required Number is 12.231

Square Root of a Fraction

To find square root of a fraction, we have to find the square roots of numerators and denominators, separately.

Ex.7 $\sqrt{\frac{2704}{81}} = ?$

Sol.

$$\begin{array}{r|l} & 52 \\ \hline 5 & 27\,04 \\ & 25\ \downarrow \\ \hline 102 & 204 \\ & 204 \\ \hline & 0 \end{array}$$

$$\sqrt{\frac{2704}{81}} = \frac{\sqrt{2704}}{\sqrt{81}} = \frac{52}{9}$$

Ex.8 Find the square root of $\frac{461}{8}$

Sol.

$$\sqrt{\frac{461}{8}} = \sqrt{\frac{461 \times 2}{8 \times 2}} = \sqrt{\frac{922}{16}} = \frac{30.3644}{4} = 7.5911 \text{ approx}$$

Note: Sometimes, numerator and denominator are not a complete square. In these types of cases, it is better to convert the given fraction into decimal fraction to find the square root.

Ex.9 $\sqrt{\frac{9261}{8400}} = ?$

Sol.

$$\sqrt{\frac{9261}{8400}} = \sqrt{1.1025} = 1.05$$

5.5 Fast Track Formulae to solve the QUESTIONS

Formula 1

If in a given number, the total number of digits are n and if n is even, then square root of that number will have $\frac{n}{2}$ digits and if n is odd, then number of digits will be $\frac{n+1}{2}$.

Ex. 10 How many digits are there in square root of 1838736.

Sol. Since, the total number of digits are 7 and 7 is odd number.

$$\therefore \quad \text{Number of digits} = \frac{7+1}{2} = 4 \text{ digits}$$

Formula 2

If any number has 5 in unit's place, then its square can be calculated as

$$(\mathscr{A}.5)^2 = \mathscr{A}(\times\mathscr{A}+1).25$$

Note: Student should understand the dot(.)just as a separation between numbers

Ex. 11 Find the square of 125?

Sol.

$$125^2 = (12\times(12+1)).25 = 15625$$

Formula 3

$$\sqrt{x\sqrt{x\sqrt{x\ldots}}}(n \text{ times}) = x^{\frac{2^n-1}{2^n}}$$

Ex. 12 Find the value of $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}}$

Sol. Here, value of $n = 5$

$$\therefore \sqrt{2\sqrt{2\sqrt{2\sqrt{2\sqrt{2}}}}} = 2^{\frac{2^5-1}{2^5}}$$
$$= 2^{\frac{32-1}{32}} = 2^{\frac{31}{32}}.$$

Formula 4

$$\sqrt{x\sqrt{x\sqrt{x\sqrt{x\ldots}}}}(\infty \text{ times}) = x$$

Ex.13 Find the value of $\sqrt{7\sqrt{7\sqrt{7\ldots}}}(\infty \text{ times}) =?$

Sol. As the expression continues to infinity, so $\sqrt{7\sqrt{7\sqrt{7\ldots}}}(\infty \text{ times}) = 7$

Formula 5

To find the value of $\sqrt{x+\sqrt{x+\sqrt{x+\ldots}}}$, find the factors of x, such that the difference between the factors is 1, then the larger factor will be the result.

Ex.14 Find the value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$
Sol. The factors of 12 with difference of 1 are 3 and 4. Here, 4 is the larger number

$$\therefore \qquad \sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}=4$$

Formula 6

To find the value of $\sqrt{x-\sqrt{x-\sqrt{x-\ldots}}}$, find the factors of x, such that the difference between the factors is 1, then the smaller factor will be the result.

Ex.15 Find the value of $\sqrt{30-\sqrt{30-\sqrt{30-\ldots}}}$
Sol. The factors of 30 with difference of 1 are 5 and 6. Here, 5 is the smaller number

$$\therefore \sqrt{30-\sqrt{30-\sqrt{30-\ldots}}}=5$$

5.6 Fast Track Techniques to solve the QUESTIONS

Technique 1

If any number is in $\frac{1}{\sqrt{a}\pm\sqrt{b}}$, then we multiply by its rationalisation factor $\sqrt{a}\mp\sqrt{b}$ in both numerator and denominator.

Ex.16 If $\sqrt{2}=1.414$, then find the value of $\frac{1}{\sqrt{9}-\sqrt{8}}$.
Sol.

$$\frac{1}{\sqrt{9}-\sqrt{8}}=\frac{1}{\sqrt{9}-\sqrt{8}}\frac{\sqrt{9}+\sqrt{8}}{\sqrt{9}+\sqrt{8}}=\frac{\sqrt{9}+\sqrt{8}}{\sqrt{9}^2-\sqrt{8}^2}=\frac{\sqrt{9}+\sqrt{8}}{9-8}$$
$$=\frac{3+2\sqrt{2}}{1}=3+2\sqrt{2}=3+2(1.414)=3+2.828=5.828$$

Technique 2

To find smallest and largest n-digit number which is a perfect square. Ex. 17 Find the biggest 5-digit number which is a perfect square.

Sol. The biggest 5-digit number = 99999

Clearly, the biggest 5-digit number which is a perfect square is less by 143.

$\therefore$ The Number $= 99999-143=99856$

	316
3	9 99 99
	9 ↓
61	99
	61 ↓
626	3899
	3756
	143

	32
3	10 00
	9 ↓
62	100
	62
	−24

Ex. 18 Find the smallest 4-digit number which is a perfect square.

Sol. Smallest 4-digit number $= 1000$

Clearly, the smallest 4-digit number which is a perfect square, will be greater by 24.

$\therefore$ The Number $= 1000 + 24 = 1024$

5.7 Cube

If a number is multiplied two times with itself, then the result of this multiplication is called the cube of that number.
For example:

- Cube of $6 = 6^3 = 6 \times 6 \times 6 = 216$
- Cube of $8 = 8^3 = 8 \times 8 \times 8 = 512$

5.7.1 Methods to Find Cube

Different methods to calculate the cube of a number are as follows

Algebraic Method

To calculate cube by this method, two formulae are used.

★ $(a+b)^3 = a^3 + 3ab(a+b) + b^3$

★ $(a-b)^3 = a^3 - 3ab(a-b) - b^3$

For example: The cube of 16 is

$$(16)^3 = (10+6)^3 = (10)^3 + 3 \times 10 \times 6(10+6) + (6)^3 = 1000 + 2880 + 216 = 4096$$

Shortcut Method

Step I The answer consists of 4 parts each of which has to be calculated separately,

Step II First write down the cube of ten's digit to the extreme left. Write the next two terms to the ricrht of it bv creatine; GP (Geometric Progression) having common ratio which is equal to $\frac{\text{unit's digit}}{\text{ten's digit}}$ and the fourth number will be cube of units digit.

Step III Write the double of 2nd and 3rd number below them.

Step IV Now, add the number with numbers written below it and write the unit's place digit in a straight line and remaining number is carried forward to the next number.

Ex. 19 Find the cube of 35.
Sol. Here, unit's digit is 5 and ten's digit is 3.

Step I Write the cube of ten's digit at extreme left *i.e.,* $3^3 = 27$

Step II Now, the next two terms on the right will be in a GP of common ratio equals $\frac{\text{unit's digit}}{\text{ten's digit}}$
So, the two terms will be $27 \times \frac{5}{3} = 45; 45 \times \frac{5}{3} = 75$ and the last term will be cube of units digit, *i.e,* $5^3 = 125$
So they are arranged as 27 45 75 125

Step III Twice the second and third terms are written under it and are added. So carry

$$\begin{array}{rr} & 152\,312 \\ & 274\,575 \\ + & 90\,150 \\ \hline & 42\,875 \end{array}$$

5.8 Cube Root

The cube root of a given number is the number whose cube is the given number. The cube root is denoted by the sign $'\sqrt[3]{}'$.
For example:

$$\text{(i) } \sqrt[3]{8} = \sqrt[3]{2 \times 2 \times 2} = 2 \quad \text{(ii) } \sqrt[3]{512} = \sqrt[3]{8 \times 8 \times 8} = 8$$

Methods to Find Cube Root

Method to calculate the Cube root of a number is as follow

Prime Factorisation Method

This method has following steps.

Step I Express the given number as the product of prime factors.

Step II Arrange the factors in a group of three of same prime numbers.

Step III Take the product of these prime factors picking one out of every group (group of three) of the same primes.

This product gives us the cube root of given number.

Ex. 20 Find the cube root of 9261

Sol. Prime factors of $9261 = (3 \times 3 \times 3) \times (7 \times 7 \times 7)$

$$\Longrightarrow \qquad \sqrt[3]{9261} = \sqrt[3]{(3 \times 3 \times 3) \times (7 \times 7 \times 7)}$$

Now taking one number from each group of three, we get

$$\sqrt[3]{9261} = 3 \times 7 = 21$$

3	9261
3	3087
3	1029
7	343
7	49
7	7
	1

5.9 Properties of Cubes and Cube Roots

- If the cube of a number is of 2 or 3-digits, then its cube root will be of 1-digit
- If the cube of a number is of 4, 5 or 6 digits, then its cube root will be of 2 digits.
- If the cube of a number have 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 in its unit's place, then its cube root will have 0, 1, 8, 7, 4, 5, 6, 3, 2 or 9 in their unit's place, respectively.
- There are only three numbers whose cube is equal to the number
  i.e., $0^3 = 0 \quad ; \quad 1^3 = 1 \quad ; \quad (-1)^3 = -1$

Note:

➪ If $\frac{p}{q}$ is a fraction, then $\sqrt[3]{\frac{p}{q}} = \frac{\sqrt[3]{p}}{\sqrt[3]{q}}$, where p and q are integers.

➪ If p is an integer, then $\sqrt[3]{-p} = -\sqrt[3]{p}$

Ex. 21 Find the value of $\sqrt[3]{\frac{0.000729}{0.085184}}$

Sol.

$$\sqrt[3]{\frac{0.000729}{0.085184}} = \sqrt[3]{\frac{729}{85184}} = \sqrt[3]{\frac{9\times 9\times 9}{44\times 44\times 44}} = \frac{9}{44}$$

Ex. 22 Find the cube root of -5832.

Sol.

$$\sqrt[3]{-5832} = -\sqrt[3]{5832} = -\sqrt[3]{18\times 18\times 18} = -18$$

Ex. 23 Find the cube root of -17576.

Sol.

$$\sqrt[3]{-17576} = -\sqrt[3]{17576} = -\sqrt[3]{26\times 26\times 26} = -26$$